

PURE Perforating

April 2003

What is PURE?

The PURE*, “Perforating for Ultimate Reservoir Exploitation”, system is a technique to obtain a dynamic under balance during the perforation process. This system is a combination of perforating gun and completion design.

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The PURE system is a propriety property of Schlumberger and is the subject of US Patent Application No.:US 2002/0020535 A1 dated February 21 2001, and PCT Application No.: WO 01/65060.

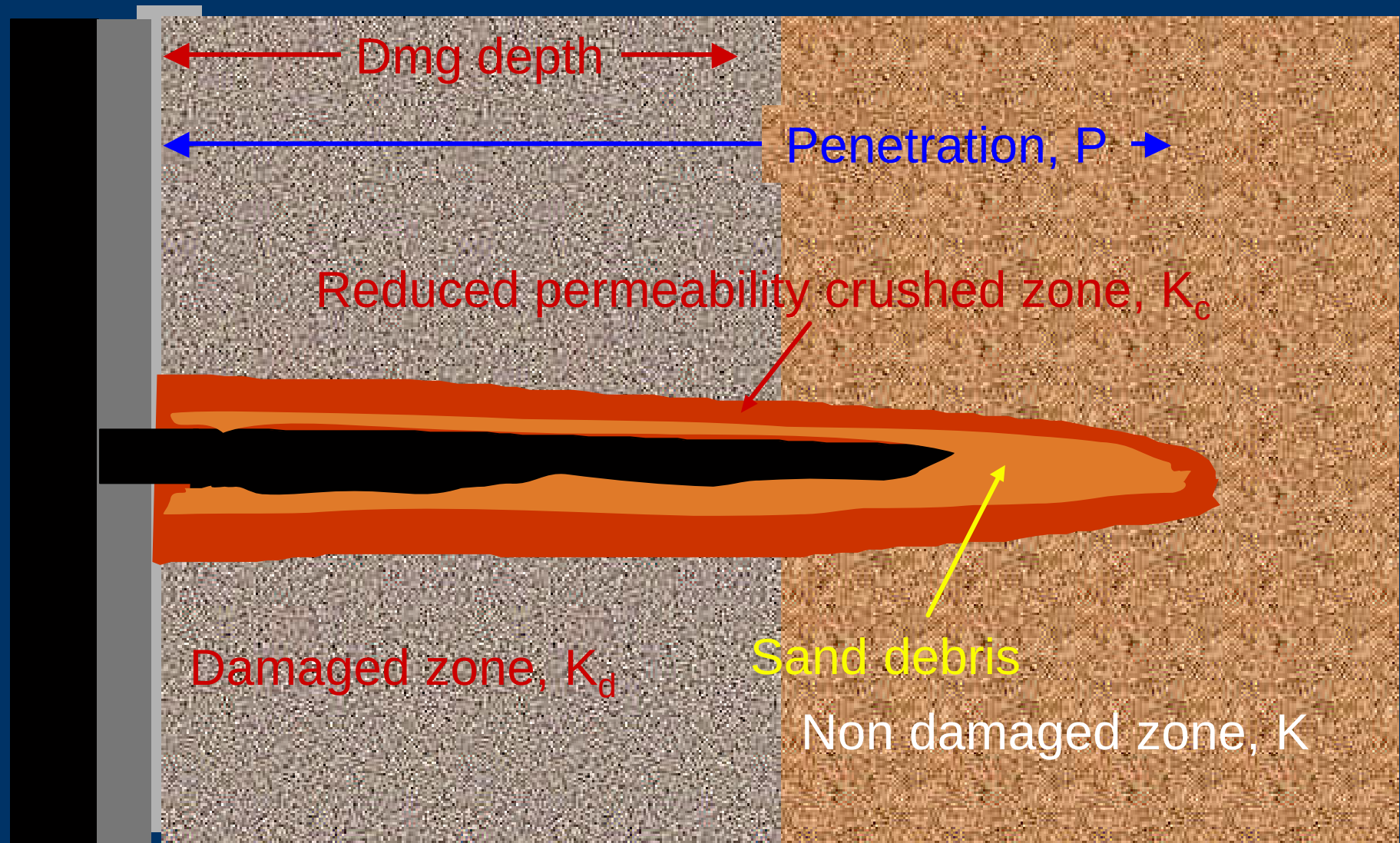
The PURE system may only be used pursuant to a contract agreed with Schlumberger and must not be distributed or divulged to third parties, or duplicated in whole or in part, without prior written permission by Schlumberger.

* Mark of Schlumberger

PURE Objectives

- Maximize well productivity or injectivity by minimizing perforation skin.
- Eliminate acid stimulation in low (>0.5 md) permeability gas wells.
- Minimize cement-to-sand face micro annulus.

Productivity = f(perforation process, ++)



Current Paradigm

- Perforating under balance is the technique of choice for maximizing productivity.
(initial wellbore pressure $<$ pore pressure)
- Over balanced perforating is damaging.
(initial wellbore pressure $>$ pore pressure)

Crushed Zone Damage

Perforating under balance helps remove crushed zone damage.

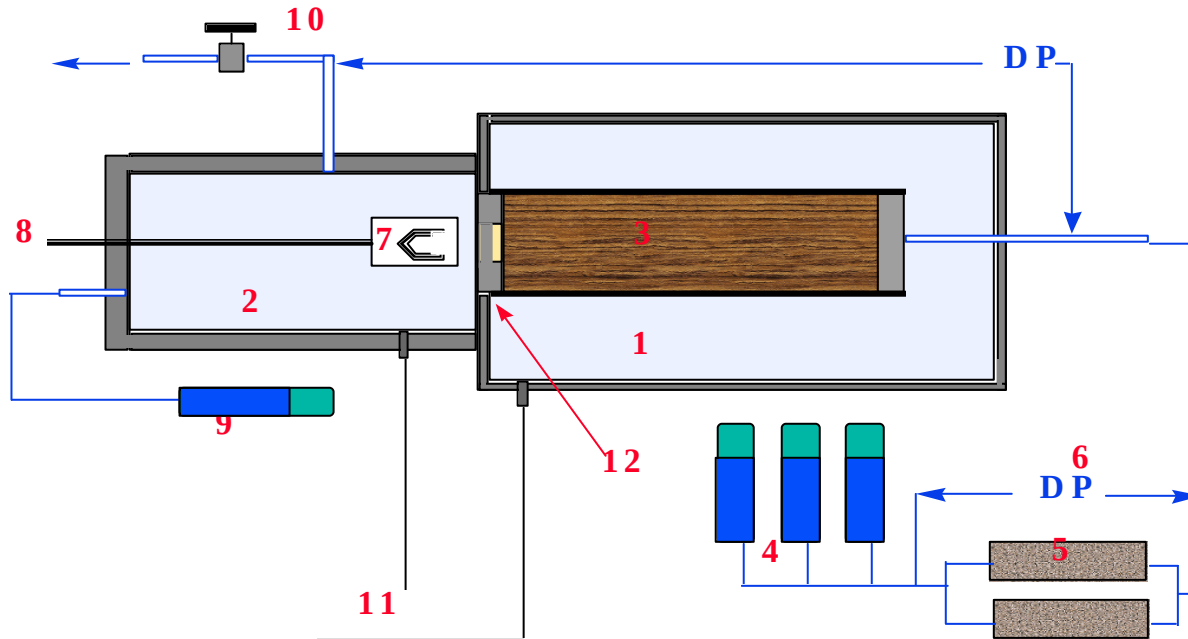
- How much under balance do we need?

Depends on rock properties (permeability, strength, porosity), effective stress and tunnel diameter and maybe on rate of drawdown.

- How much under balance do we get?

Depends on wellbore geometry etc.

Single-shot Tests

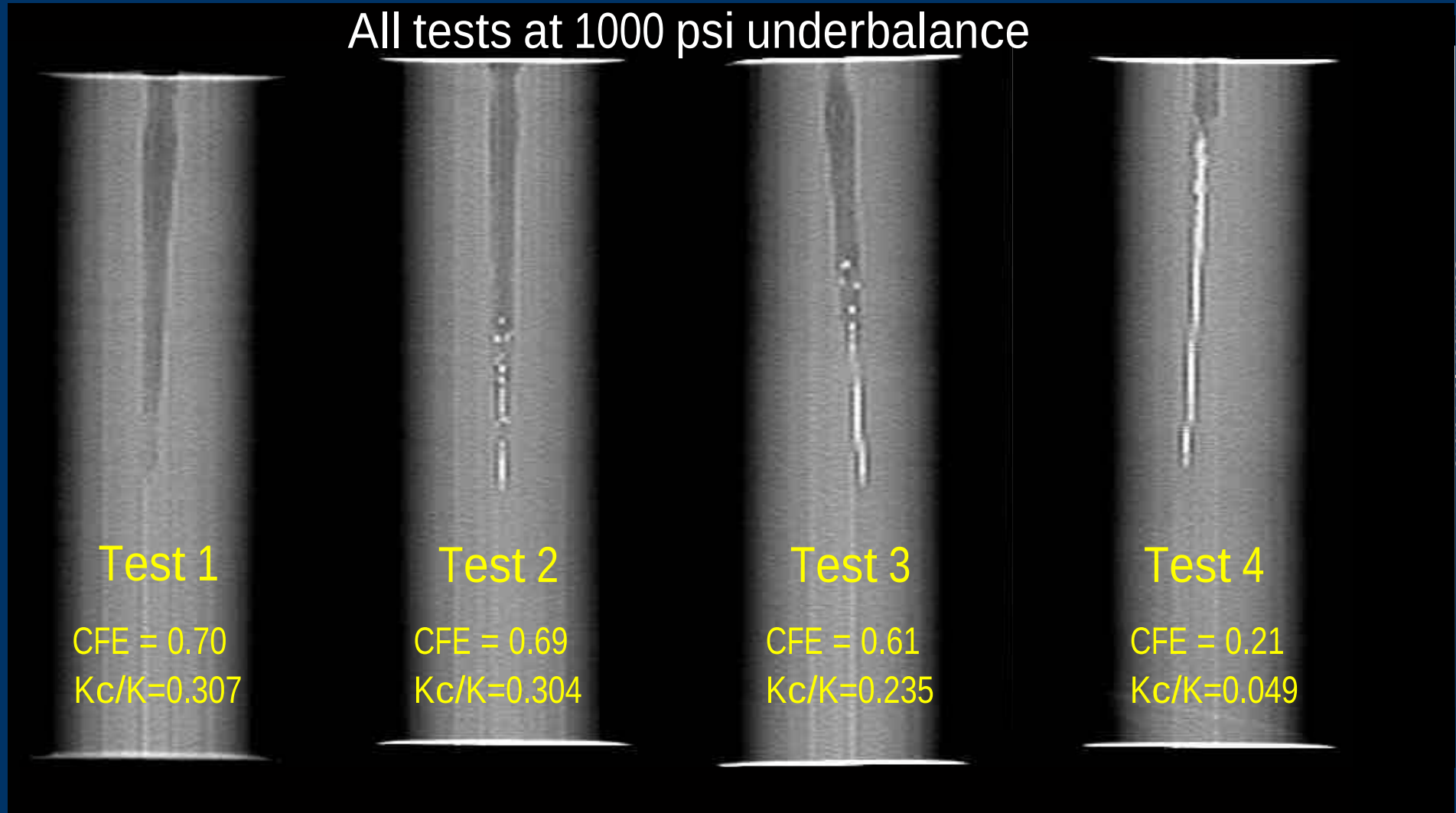


Experimental set-up

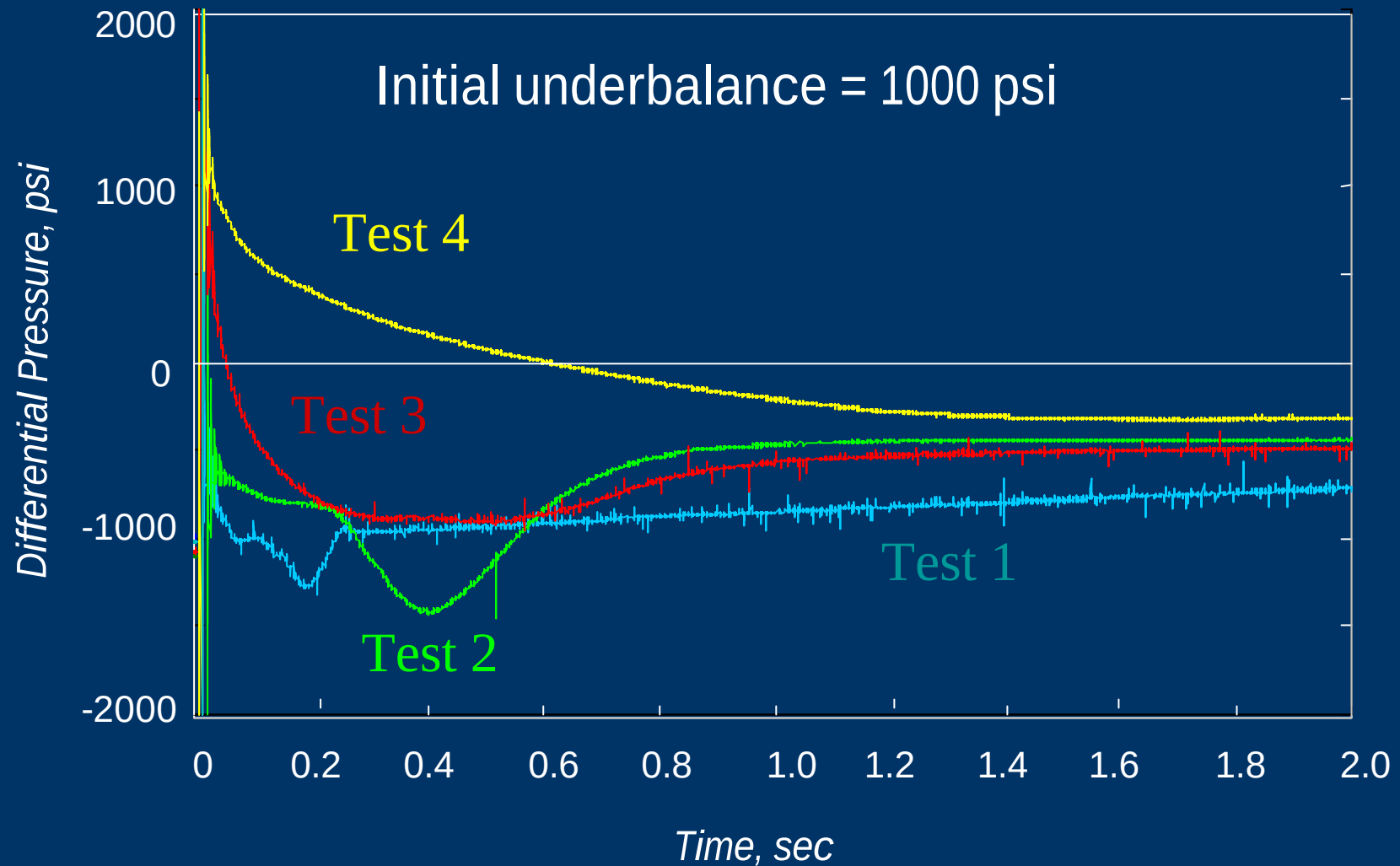
1. Confining chamber with confining fluid (Kerosene)
2. Simulated wellbore with wellbore fluid
3. Core sample with pore pressure and pore fluid
4. 30 gallon accumulator with pre-determined gas pre-charge
5. Simulated reservoir rock samples
6. Differential pressure gauges
7. Gun with the shaped charge
8. Shooting leads
9. 5 gallon accumulator
10. Micrometer valve
11. PCB gauges
12. Shooting Plate

Choose The Best Perforation?

All tests at 1000 psi underbalance



Perforation Dynamics

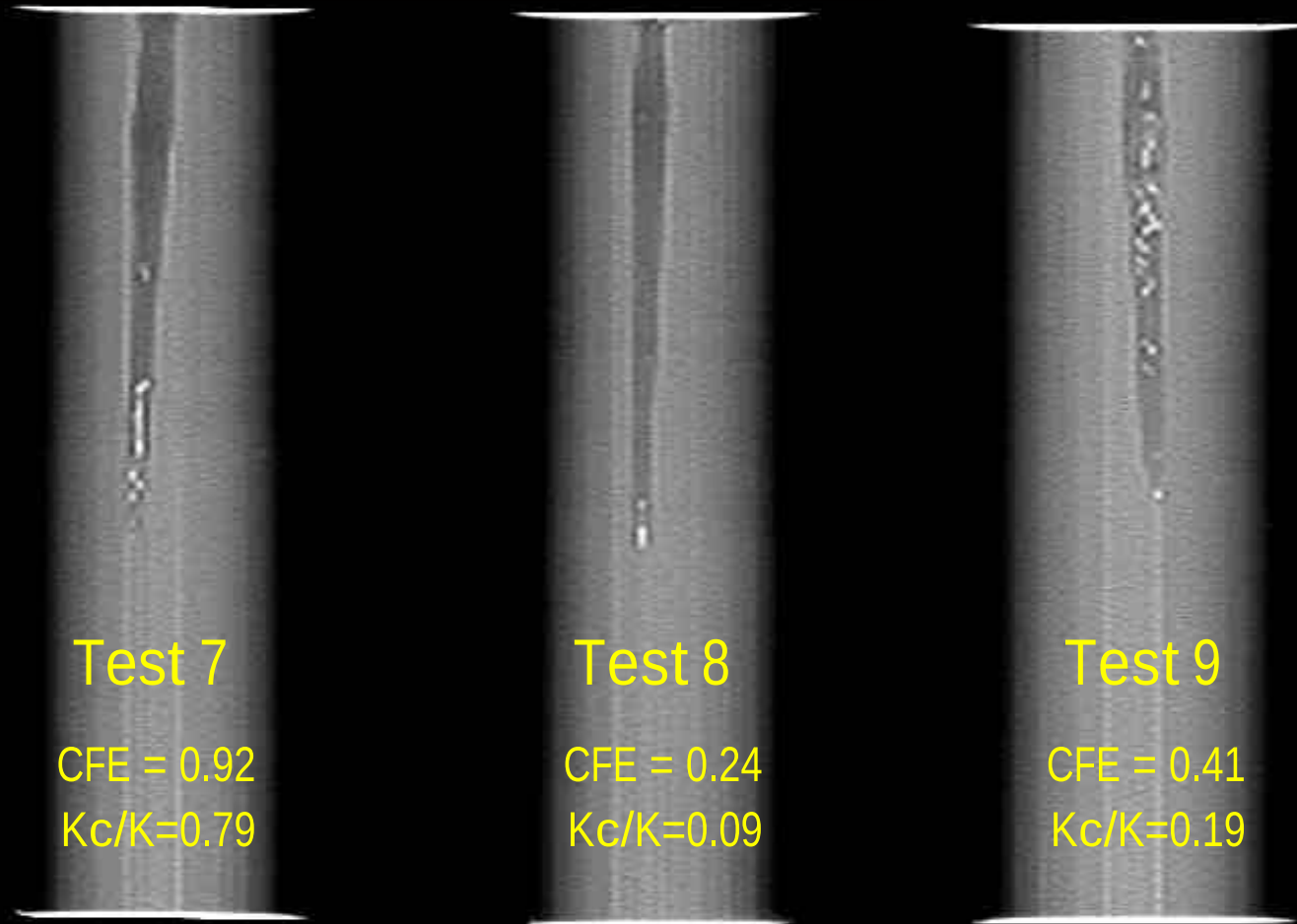


Static Underbalance Does Not Provide Sufficient Information

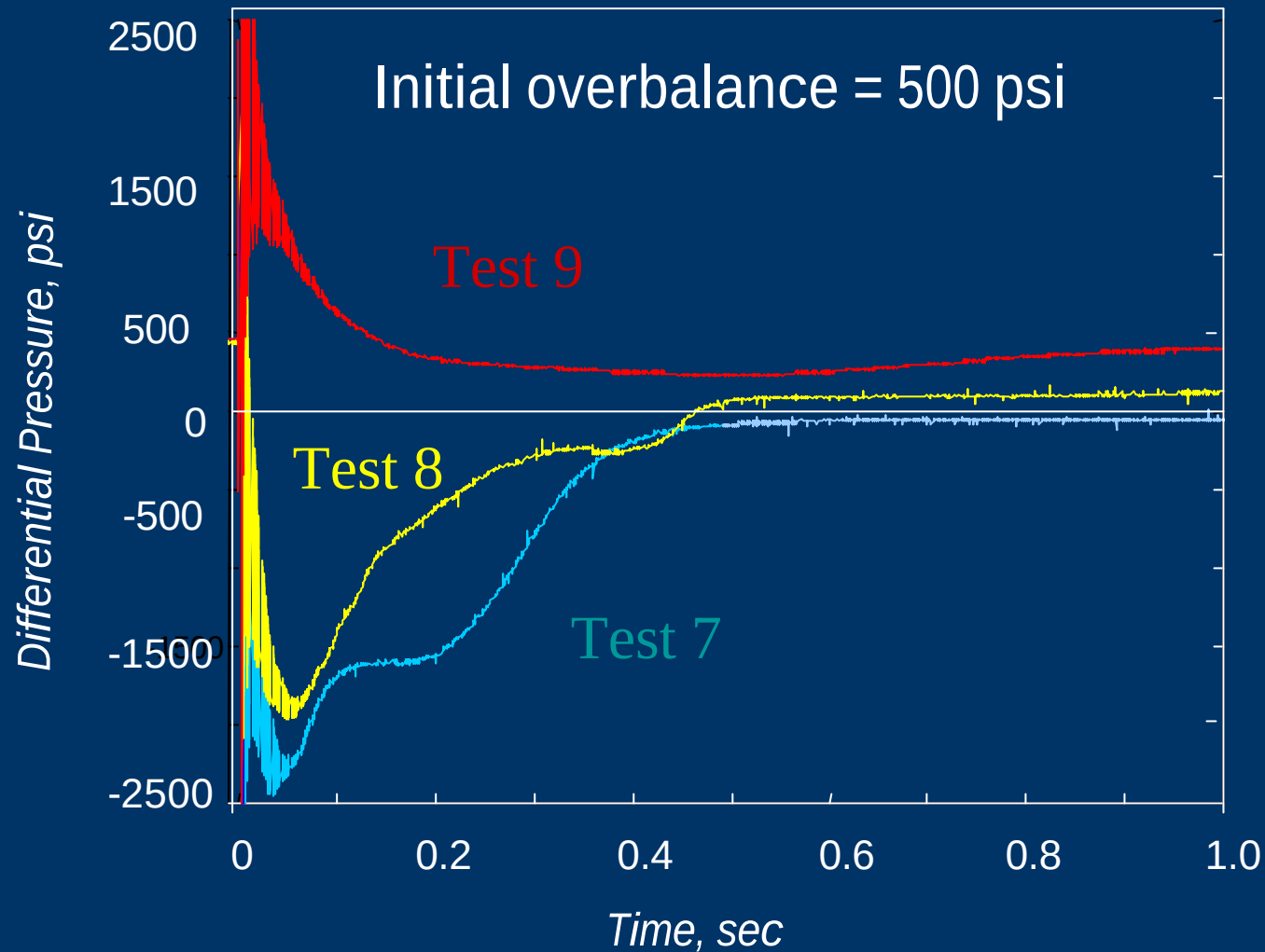
- All four tests show different pressure behaviour with time
- Tests 1 & 2 remain underbalanced throughout
- Tests 3 & 4 go overbalanced after firing and take some time to go back underbalanced
- Test 4 takes the longest time to go underbalanced

Choose The Best Perforation?

Tests shot at 500 psi overbalance



Perforation Dynamics



Overbalanced Perforating May also Provide Clean Perforations

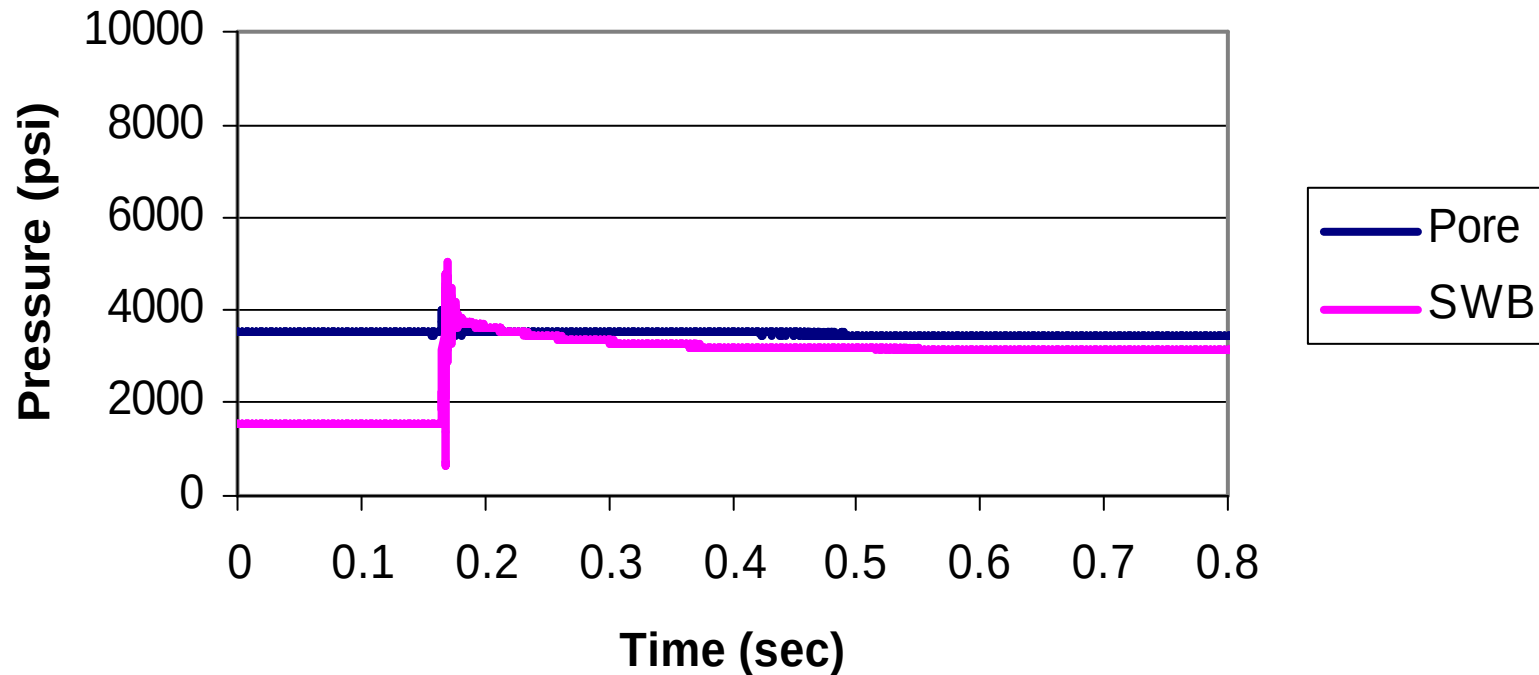
- Test 7 & 8 show a dramatic dynamic underbalance—2000 to 2500 psi
- Test 7 remains underbalanced throughout and gave a better CFE than the underbalanced tests
- Test 9 goes to 2500 psi overbalanced and remains overbalanced
- Test 8 suddenly goes overbalanced at 450 msec—a water-hammer effect plugging off the perforation tunnel

Lab Tests

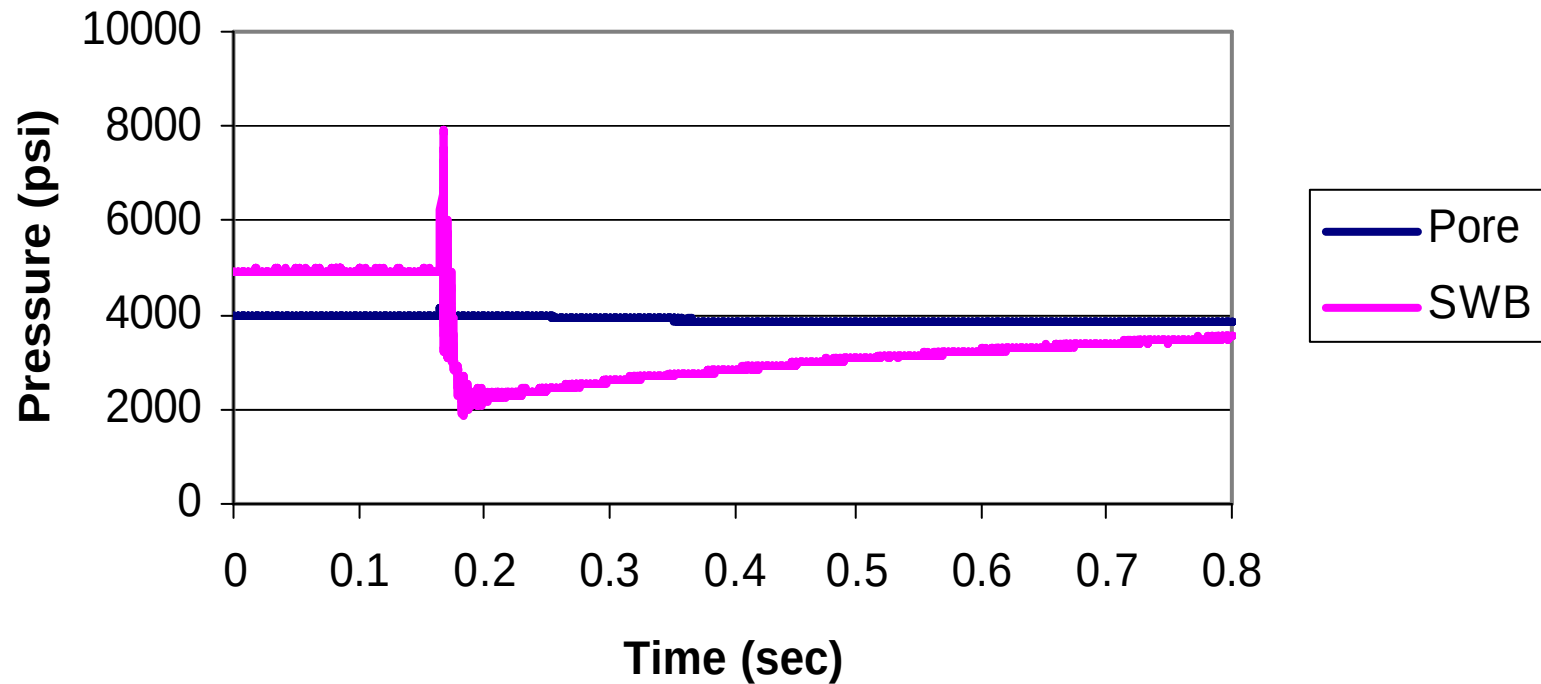
Gas @ Residual Brine Saturation

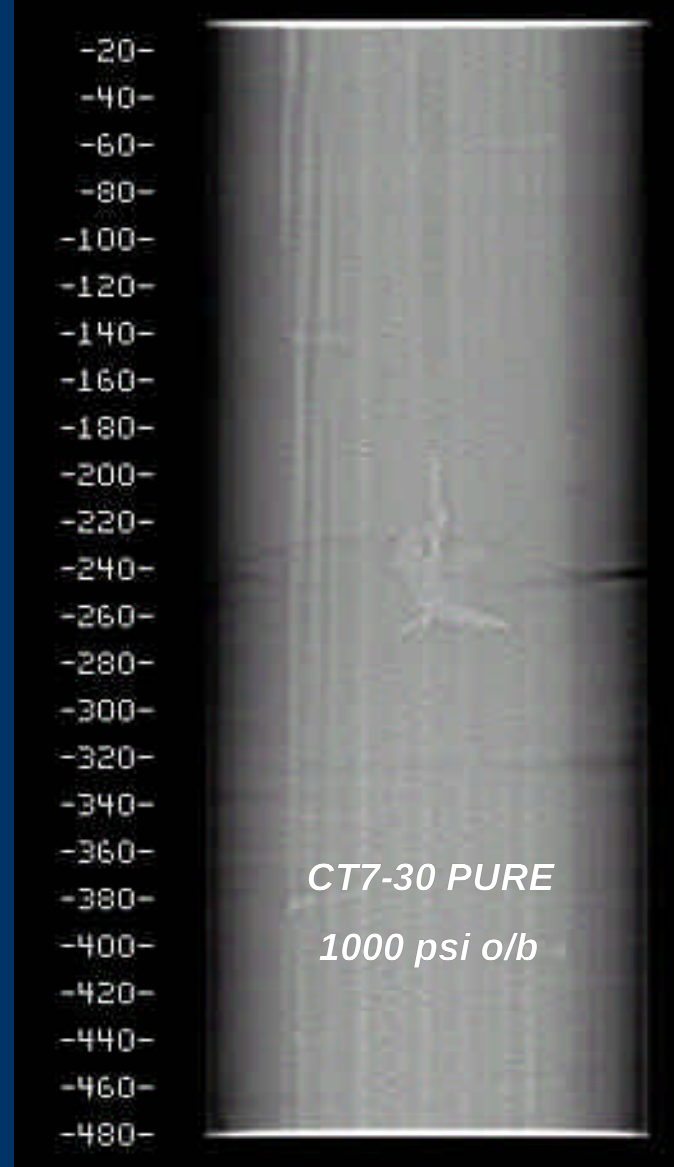
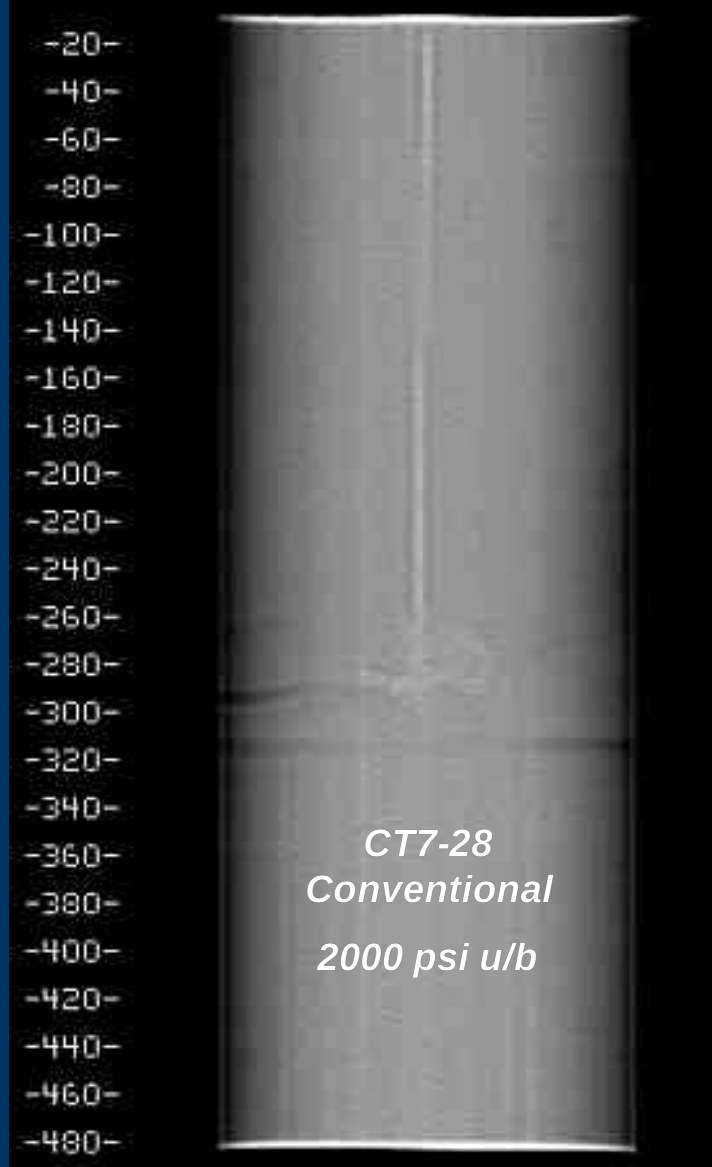
-1000 psi overbalance	500 psi underbalance	1000 psi underbalance	1500 psi underbalance	2000 psi underbalance	-1000 psi overbalance PURE™
CFE = 0.59	CFE = 0.5	CFE = 0.56	CFE = 0.61	CFE = 0.64	CFE = >1

CT7-28 (Conventional @ 2000 psi u/b)



CT7-30 (PURE @ 1000 psi o/b)





Static Wellbore Conditions Do Not Provide Sufficient Information

- Post perforation wellbore dynamics critical.
- What magnitude and rate of transient underbalance determine perforation cleanup.
- Need to identify and then control the parameters that influence the dynamics.

What are PURE system features?

- Maximizes the transient dynamic under balance during the first few 100 milliseconds.
- Allows independent control of the post-perforation surge (flow during seconds after perforating).
- Reduces dynamic wellbore pressure.
- Does not require high initial (static) under balance.
- Can start with an initial overbalance with a TCP string.

What are PURE system benefits? —1

- Increases productivity or injectivity by: maximizing perforation cleanup.
- Effectively treats all the perforated intervals.
 - a. More effectively cleanups up a wide range of permeabilities.
 - b. Increases effective shot density.
- Reduces pressure loading on the perforating string.
- Reduces the initial under balance requirements, saving costs of N_2 , completion fluids, etc.

What are PURE system benefits? —2

- Minimizes hydraulic cement-to-sand micro-annulus.
- Minimizes the tendency to sand-in the guns by controlling the post perforation surge.
- Eliminates acid stimulation of low permeability gas and oil wells.
- Provides system tradeoff between penetration depth, shot density and perforation damage (a system approach versus API data only) .

Which Wells are PURE Candidates?

Reservoir Pressure

> 2000 psi (designs have been made for lower

Wellbore Fluid

pressures)

Must be liquid around the guns

Conveyor (SL, WL, CT, TCP, PCP)

Initial static underbalance

Any conveyor

Initial static overbalance

TCP

Field Test Overview

- Field tests to date = 42, production results on 35.
- Success rate (poor reservoirs eliminated) = 27/30
 - Success defined as better than expected results or eliminated acid stimulation.
 - One operational failure.
 - One expected fluid properties problem.
 - One had a completion restriction.
- WL (17), TCP/MXAR (16), OrientXact (9)
- Producers (32), Injectors (10)

Chevron Wyoming Painter Field

- Low permeability, high strength Nugget sandstone.
- 3-3/8 TCP guns.
- Oil and gas.
- Results, six horizontal wells: SPE 77364
 - Zero skins, eliminated \$150K acid jobs on four wells
 - Operational issues on one well, no improvement after acidizing on one well.

Weber Formation Low Permeability Gas Wells

- Permeability 0.5 – 1.5 md.
- Reservoir pressure ~3050 psi and 2300 psi
- Permanent completion; 2 7/8-in. guns
- Initial overbalance = 3000 and 3700 psi
- Results:
 - Production exceeded expectations at 5.5 mmscf/d and 3.4 mmscf/d without acid stimulation
 - All previous wells required acid stimulation to produce

GOM Monobore Oil Well

- Permanent completion, 40 ft. 4.72 HSD guns with MAXR.
- Reservoir pressure ~ 5200 psi.
- Minimum initial underbalance <500 psi.
- Nodal analysis was run assuming zero formation and perforation skin — production matched this scenario.

Caspian Sea Carbonate

- High UCS, low permeability (2 md), high H₂S
 - Laboratory Tests: Mud completion fluid
 - Conventional overbalance: skin= 25 after perforating, 3 after repeated acid flush/ injection/ production cycles.
 - PURE with 1000 psi initial underbalance (3000 psi dynamic ub), killed with mud and then produced: skin=1.4 before kill, 2.2 after cleanup (production)
- Field results
 - Perforated close to balance using PURE design with mud in hole
 - Excellent productivity

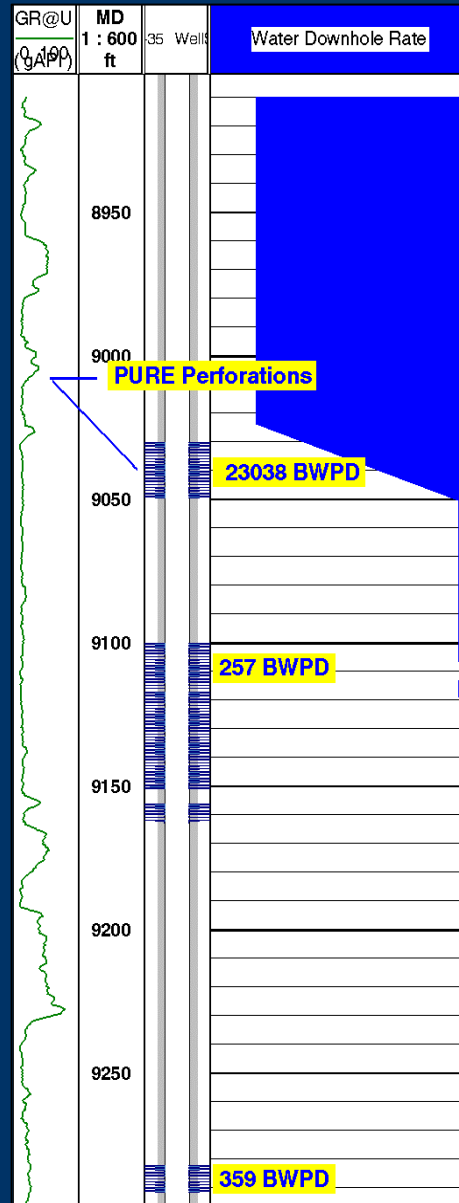
Alaska Oil Well

- Fourteen feet of perforations
- WL with initial modest UB
- PURE charges used
- Oil rate of 500 bbl/d considered success, 800 bbl/d great success
- Initial oil rate of 1400 bbl/d before gassing out; will be turned into an injector

Alaska Injector

- Added 20 ft of perforations to existing 90 ft.
- WL at balanced conditions
- PURE charges used
- Injectivity increased from 17,500 to 25, 000 bbl/d. PURE zone taking 23,038 bbl/d.

Alaska Injector Production Log



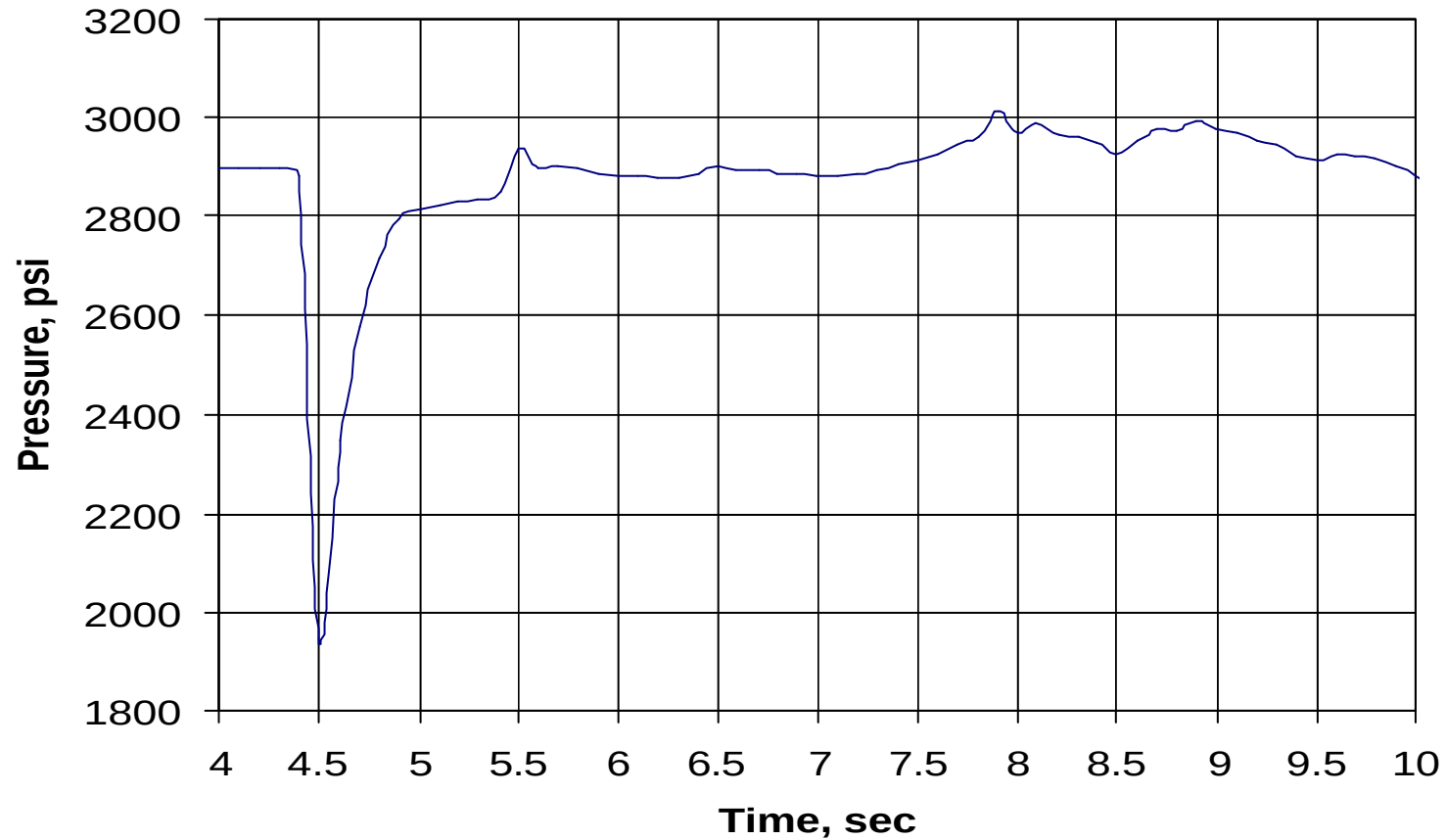
Algeria Oil Wells

- Sandstone oil wells , 5100 psi, 200 md.
- Three zones
- WL with modest initial UB (three runs, well flowed for runs 2 and 3)
- PBU skin measured at 0, typically five
- Results described as “startling”
- Two more successful wells completed

Aberdeen Injectors

- Water injector, added perforations
- Multiple WL runs
- PURE charges used
- Injection/ft increased by 70% (assumes all zones injecting at same rate/ft)
- Similar results on second injector

PURE Dynamic UB from Injector



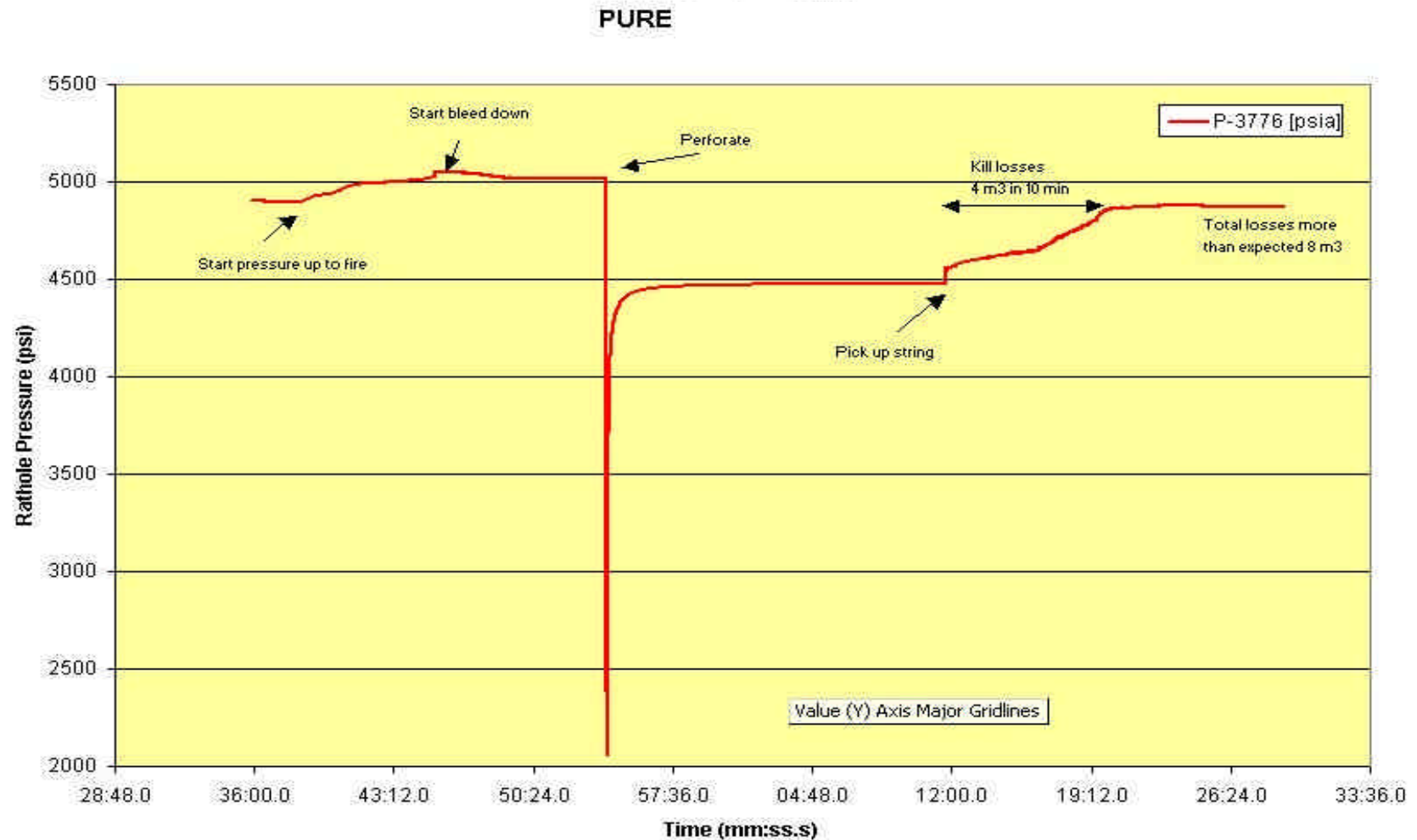
PURE @ 6 spf Versus Conventional @ 12 spf

- Gas well
- Top zone shot PURE @ 290 psi initial UB
- Lower zone shot conventional with well flowing
- PLT run:
 - PURE zone 3 times expectation
- Both zones reperforated conventional and PLT run
 - No increase in production from PURE zone
 - Large increase in production from lower zone
- Conclusions: Use of PURE @ 6 spf much more effective than conventional @ 12 spf.

Long Horizontal PURE Wells

- Ten wells, all with initial 500 psi overbalance.
- Perforations oriented to top-side for sand prevention.
- Best injector in field.
- PI ten times next best well.
- Dynamic underbalance measured.

Measured Dynamic Underbalance



Happy PURE Clients

- Chevron
- Anadarko
- Hydro
- Sonatrach
- BP
- Statoil
- Amerada Hess
- Shell
- TFE

SPE Papers

- SPE 71642
Laboratory Experiments Provide New Insights into Underbalanced Perforating
- SPE 77364
New Underbalanced Perforating Technique Increases Completion Efficiency and Eliminates Costly Acid Stimulation